

Local Bifurcations (local to equilibrium)

$$\dot{X} = V(X; \lambda) \quad X \in \mathbb{R}^n, \lambda \in \mathbb{R}^k$$

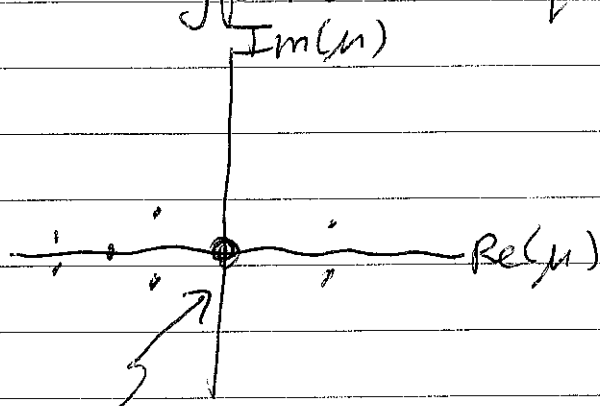
↖ parameters

equilibrium: $V(X_0; \lambda_0) = 0$
(for $\lambda = \lambda_0$)

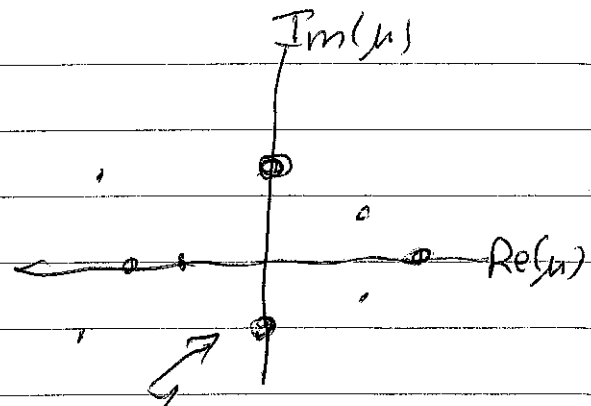
- If X_0 is hyperbolic, then small changes to parameters λ do not change $\dim(E^s)$ & $\dim(E^u)$, so stability doesn't change. Equilibrium can also not be destroyed by small change to λ (implicit function theorem argument below).

★ Hyperbolic equilibria are robust under small perturbations of eqns.

- Local bifurcations are associated with non-hyperbolic equilibrium



"steady state bifurcations"



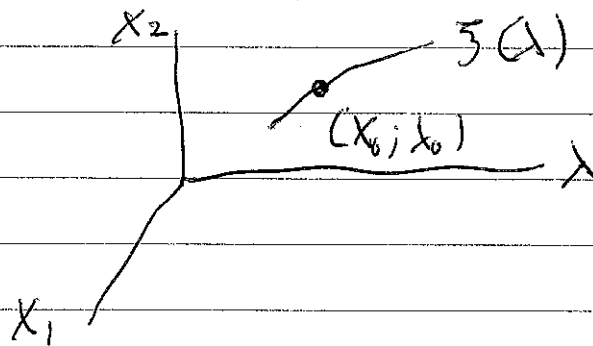
"Hopf bifurcations"

Only steady state bifurcations can lead to changes in # of equilibria

At steady state bifurcation: $\text{Det}[D_x V(x_0; \lambda)] = 0$

ie $\mu=0$ is eigenvalue of $D_x V(x_0; \lambda_0)$ at the bifurcation pt $\lambda = \lambda_0$

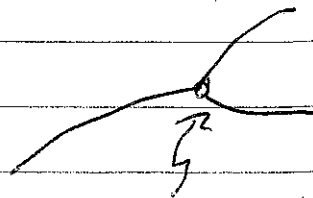
Implicit Function Thm. : If $D_x V(x_0; \lambda_0)$ is non-singular ($\text{Det} \neq 0$), then there are open sets $V \subset \mathbb{R}^n$ containing x_0 , $W \subset \mathbb{R}^k$ containing λ_0 , and a unique C^r function $\xi(\lambda) : W \rightarrow V$ for which $V(\xi(\lambda); \lambda) = 0$, $x_0 = \xi(\lambda_0)$.



Side: If $D_x V(x_0; \lambda_0)$ is non-singular then we could estimate $\xi(\lambda)$, in neighborhood of $(x_0; \lambda_0)$ as a line

$$0 = V(x; \lambda) = V(x_0; \lambda_0) + D_x V(x_0; \lambda_0)(x - x_0) + D_\lambda V(x_0; \lambda_0)(\lambda - \lambda_0) + o(x - x_0, \lambda - \lambda_0)$$

$$x \approx x_0 - [D_x V(x_0; \lambda_0)]^{-1} D_\lambda V(x_0; \lambda_0)(\lambda - \lambda_0)$$

Bifurcations:

Implicit function thm.
must fail here

$$\text{Det}(D_x V(x_0; \lambda_0)) = 0$$

Simplest steady state bifurcation:
 $\mu = 0$, with multiplicity one

Nonhyperbolic Hartman - Grobman

$$(*) \begin{cases} \dot{x} = Cx + F(x, y, z) & \text{center directions} \\ \dot{y} = Sy + G(x, y, z) & \text{stable directions} \\ \dot{z} = Uz + H(x, y, z) & \text{unstable directions} \end{cases}$$

For a C^k , $k \geq 1$ vector field with fixed pt $(0, 0, 0)$, F, G, H are $o(x, y, z)$

$\exists$ neighborhood N of $(0, 0, 0)$ s.t.

$W_{loc}^c(0, 0, 0) = \{(x, g(x), h(x)) : x \in E^c\} \cap N$
and dynamics of $(*)$ are topologically conjugate to

$$\begin{aligned} \dot{x} &= Cx + F(x, g(x), h(x)) & \leftarrow \text{gives dynamics in } W_{loc}^c \\ \dot{y} &= Sy \\ \dot{z} &= Uz \end{aligned}$$

at $x=x_0, \lambda=\lambda_0$: $\mu=0$, multiplicity one
so $\dim(W^c)=1$

Consider case where $\lambda \in \mathbb{R}$, Goal: determine behavior in W^c for λ near λ_0 , the steady state bifurcation pt.

$$\begin{cases} \dot{x} = 0 \cdot x + F(x, \lambda, \overbrace{g(x, \lambda)}^y, \overbrace{h(x, \lambda)}^z) = f(x, \lambda) \\ \dot{\lambda} = 0 \\ \dot{y} = Sy \\ \dot{z} = Uz \end{cases}$$

↑
parameter treated as
dynamical variable

2-d extended center manifold $(x, \lambda) \in \mathbb{R}^2$

$$\dot{x} = f(x, \lambda) \quad \text{on center manifold}$$

Assume bifurcation happens at $x=0, \lambda=0$:

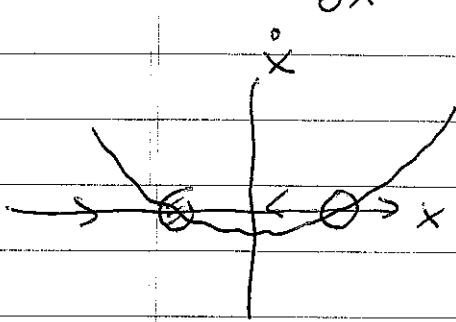
$$\left. \begin{aligned} f(0,0) &= 0 \\ \frac{\partial f}{\partial x}(0,0) &= \mu = 0 \end{aligned} \right\} \begin{array}{l} \text{Defining conditions} \\ \text{for bifurcation at } x=\lambda=0. \end{array}$$

Taylor expand $f(x, \lambda)$ in x & λ :

$$\begin{aligned} f(x, \lambda) &= \cancel{f(0,0)} + \cancel{\frac{\partial f}{\partial x}(0,0)} x + \frac{\partial f}{\partial \lambda}(0,0) \lambda \\ &\quad + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(0,0) x^2 + \frac{\partial^2 f}{\partial \lambda \partial x}(0,0) x \lambda + \frac{1}{2} \frac{\partial^2 f}{\partial \lambda^2}(0,0) \lambda^2 + \dots \end{aligned}$$

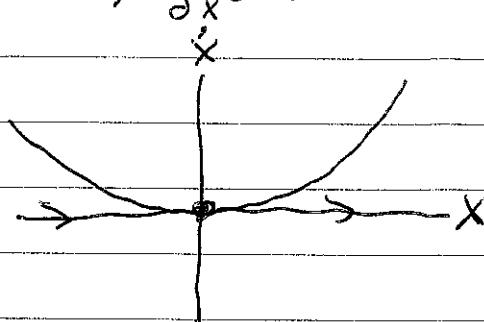
Saddle-node or fold: $\frac{\partial f}{\partial \lambda}(0,0) \neq 0, \frac{\partial^2 f}{\partial x^2}(0,0) \neq 0$

if $\frac{\partial f}{\partial \lambda}(0,0) > 0, \frac{\partial^2 f}{\partial x^2}(0,0) > 0$



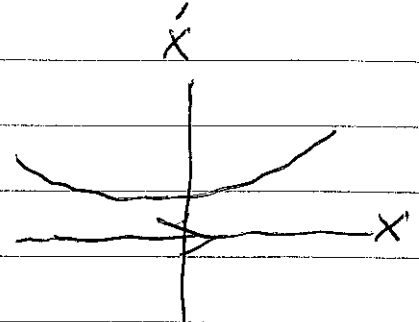
$\lambda < 0$

1 stable &
1 unstable
equilibrium



$\lambda = 0$

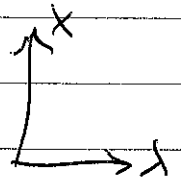
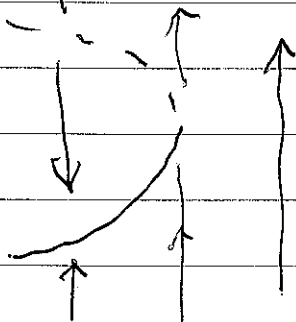
1 unstable



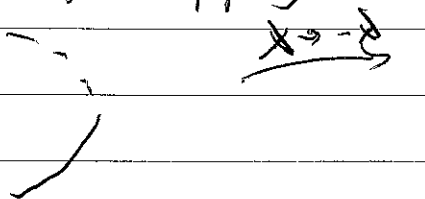
$\lambda > 0$

no equilibria

Bifurcation Diagram



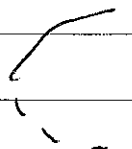
other sign choices lead to diagrams obtained by flipping about $x=0, \lambda=0$ or both



$x \rightarrow -x$



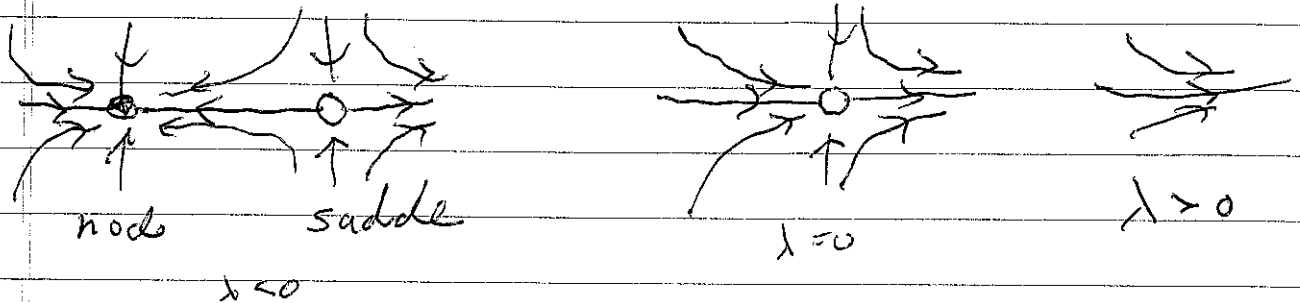
$\lambda \rightarrow -\lambda$



$x \rightarrow -x$

$\lambda \rightarrow -\lambda$

"saddle-node" terminology comes from higher dimensional cases of this, e.g. in $\mathbb{R}^2$



Transcritical bifurcation: happens naturally in problems where there is an equilibrium that exists for all parameter values, e.g. zero population state, i.e. a trivial equilibrium $x=0$

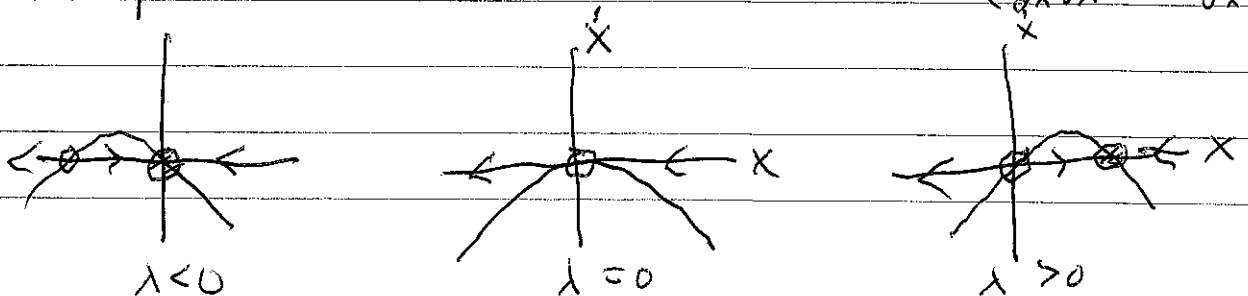
$$\Rightarrow \dot{x} = f(x, \lambda) = x g(x, \lambda)$$

$$\Rightarrow \frac{\partial f}{\partial \lambda}(0, 0) = x \frac{\partial g}{\partial \lambda} \Big|_{x=\lambda=0} = 0$$

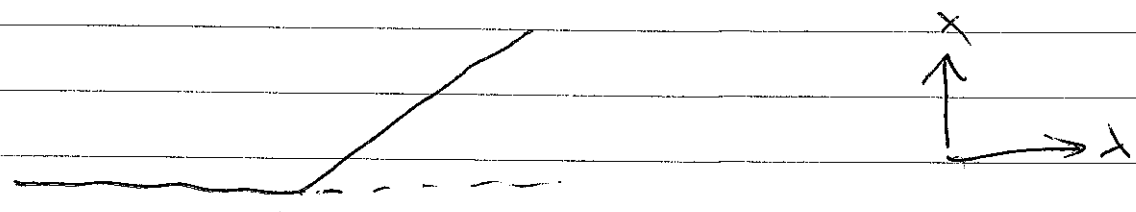
Replace $\frac{\partial f}{\partial \lambda}(0, 0) \neq 0$ by $\frac{\partial^2 f}{\partial x \partial \lambda}(0, 0) \neq 0$

assume $\frac{\partial^2 f}{\partial x^2}(0, 0) \neq 0$

example: $\dot{x} = x(\lambda - x) = x\lambda - x^2$ $\left(\frac{\partial^2 f}{\partial x \partial \lambda} > 0, \frac{\partial^2 f}{\partial x^2} < 0\right)$

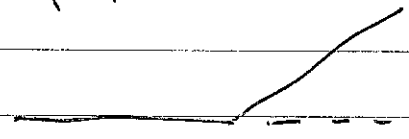


Bifurcation Diagram

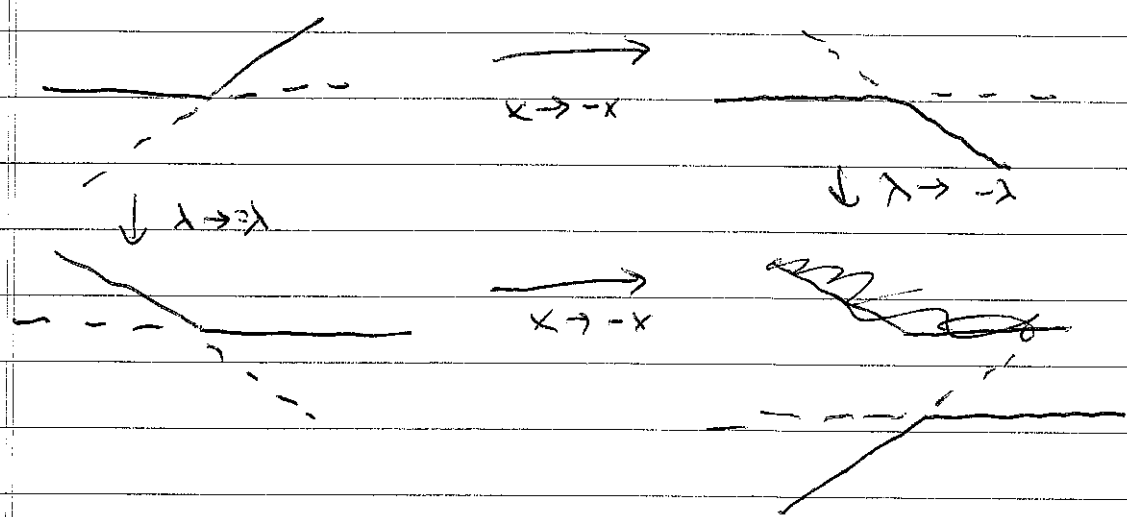


exchange of stability occurs
at the transcritical bifurcation
from the trivial equilibrium
to the non-trivial one.

If population model so $x \geq 0$



Again we can find other sign choices
b) flipping in x & λ .



Final Example: Pitchfork bifurcation occurs naturally in problems with a reflection symmetry $x \rightarrow -x$

If $\tilde{x}(t)$ is a soln of $\dot{x} = f(x)$, then so is $-\tilde{x}(t)$.

$$\dot{\tilde{x}} = f(\tilde{x})$$

$$(-\tilde{x})' = f(-\tilde{x})$$

$$-f(\tilde{x}) = f(-\tilde{x})$$

" f is $\mathbb{Z}_2$ -equivariant"

f is odd in x

$$\dot{x} = x g(x^2, \lambda)$$

$$\Rightarrow \frac{\partial f}{\partial \lambda}(0,0) = 0, \quad \frac{\partial^2 f}{\partial x^2}(0,0) = 0 \quad \text{by symmetry}$$

Two distinct cases to consider here

$$\frac{\partial^3 f}{\partial x^3}(0,0) > 0 \quad \& \quad \frac{\partial^3 f}{\partial x^3}(0,0) < 0$$

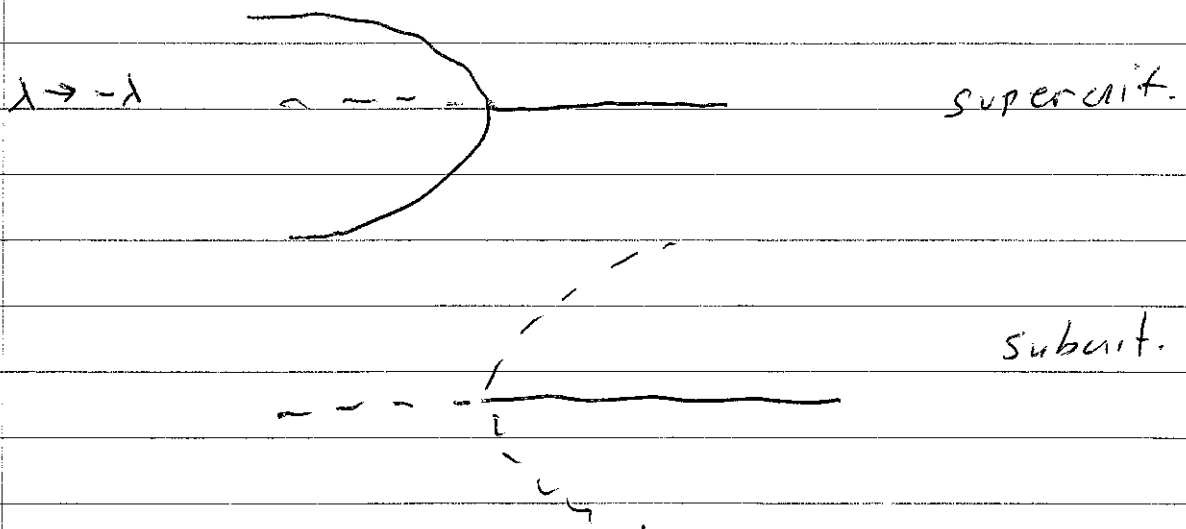
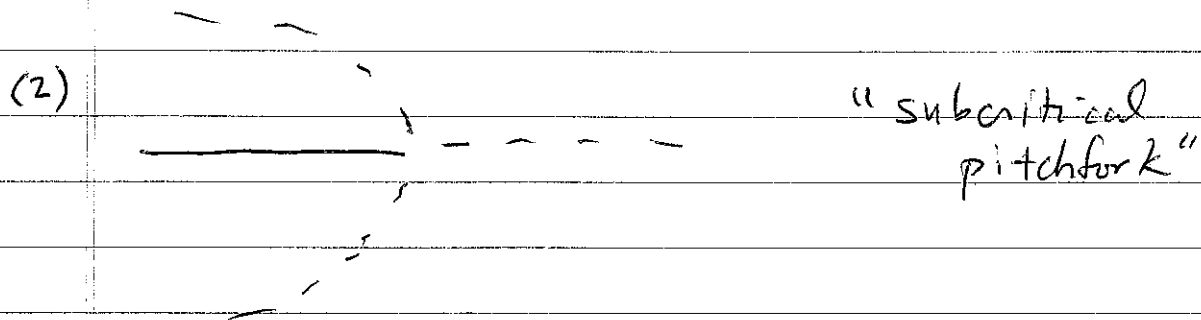
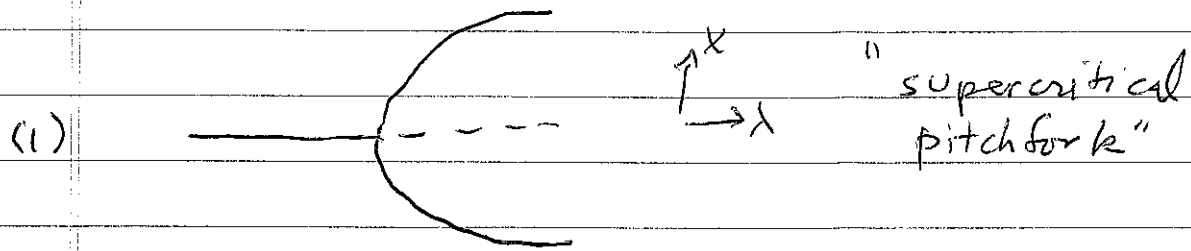
examples

$$(1) \quad \dot{x} = \lambda x - x^3$$

$$(2) \quad \dot{x} = \lambda x + x^3$$

} can't use $x \rightarrow -x$
to turn (1) into
(2) or (2) into (1)
since symmetric under
 $x \rightarrow -x$

Bifurcation Diagrams



~~This bifurcation demonstra~~

This is simplest "symmetry-breaking bifurcation"